

YEAR 7 • UNIT 2 • EXTRA HOMEWORK

Split, Multiply, Add

Two-digit multiplication • dividing • and the same grid with a letter in it

Name: _____

Class: _____

Date given: _____

Date due: **Monday 21 September 2026****Why this sheet exists**

Nobody multiplies by 16. They **split it up**: 4×10 is 40, 4×6 is 24, and $40 + 24 = 64$.

That is three steps — **split, multiply, add** — and it is also, exactly, what expanding a bracket is. 4×16 and $4(x + 3)$ are drawn in the **same grid**. Part 1 of this sheet practises the grid on numbers. Part 4 puts a letter in one box. Nothing else changes.

No calculator. Write in every box.

Part 1 • Split it up

F1 Tens and ones

Split each number. The first one is done for you.

(a) $47 = 40 + 7$

(d) $18 = \underline{\hspace{2cm}}$

(g) $34 = \underline{\hspace{2cm}}$

(b) $23 = \underline{\hspace{2cm}}$

(e) $52 = \underline{\hspace{2cm}}$

(h) $70 = \underline{\hspace{2cm}}$

(c) $65 = \underline{\hspace{2cm}}$

(f) $96 = \underline{\hspace{2cm}}$

(i) $89 = \underline{\hspace{2cm}}$

F2 Times a ten

This is the box that people get wrong, so do a page of them first.

Multiply, then put the zero back on

$4 \times 30: 4 \times 3 = 12, \text{ so } 4 \times 30 = 120.$

$20 \times 30: 2 \times 3 = 6, \text{ two zeros, so } 20 \times 30 = 600.$

(a) $4 \times 30 = \underline{\hspace{2cm}}$

(d) $9 \times 60 = \underline{\hspace{2cm}}$

(g) $20 \times 30 = \underline{\hspace{2cm}}$

(j) $30 \times 80 = \underline{\hspace{2cm}}$

(b) $6 \times 50 = \underline{\hspace{2cm}}$

(e) $8 \times 20 = \underline{\hspace{2cm}}$

(h) $40 \times 50 = \underline{\hspace{2cm}}$

(k) $50 \times 50 = \underline{\hspace{2cm}}$

(c) $7 \times 40 = \underline{\hspace{2cm}}$

(f) $3 \times 90 = \underline{\hspace{2cm}}$

(i) $60 \times 70 = \underline{\hspace{2cm}}$

(l) $20 \times 90 = \underline{\hspace{2cm}}$

F3 One row: two digits by one digit

Fill the two boxes, then add them.

The grid, once, in full 4×16 means $4 \times (10 + 6)$. One box, one multiplication.

$\times$	10	6
4	40	24

$40 + 24 = 64$

Split 16 into $10 + 6$. **Multiply** each box. **Add** the boxes.

$\times$	20	3
7		

$$\text{(a) } 7 \times 23$$

$$= \underline{\hspace{2cm}}$$

$\times$	40	8
6		

$$\text{(b) } 6 \times 48$$

$$= \underline{\hspace{2cm}}$$

$\times$	50	4
8		

$$\text{(c) } 8 \times 54$$

$$= \underline{\hspace{2cm}}$$

$\times$	30	7
9		

$$\text{(d) } 9 \times 37$$

$$= \underline{\hspace{2cm}}$$

F4 Now without the printed grid

Draw your own grid in the box, then write the answer under it.

6×34	8×72	7×85	9×46
5×68	4×93	8×57	7×64
9×78	6×95	7×49	8×86

Part 2 · Two digits by two digits**F5 Four boxes instead of two**Both numbers get split, so the grid has **four** boxes.

23×14 , once, in full

×	20	3
10	200	30
4	80	12

$$200 + 30 + 80 + 12 = 322$$

Four boxes, four multiplications, then add all four.

Three boxes means one was missed, and the answer comes out far too small.

×	30	2
10		
5		

(a) 32×15
= _____

×	40	6
20		
3		

(b) 46×23
= _____

×	50	7
30		
4		

(c) 57×34
= _____

×	60	8
40		
2		

(d) 68×42
= _____

×	70	4
20		
6		

(e) 74×26
= _____

×	80	5
30		
7		

(f) 85×37
= _____

F6 Draw the grid yourself

Twelve of them. Draw a four-box grid in each square.

24×13	35×21	47×16
52×34	63×27	78×45
86×53	94×38	75×64

29×37	56×48	83×26
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F7 Use it**Word help – the words that mean multiply**

each · every · per one of them costs this much, so all of them cost this much times the number

altogether · in total the whole lot added up

rows of · boxes of this many groups, each the same size

1. Mr Bowen buys **24** boxes of pencils. Every box holds **36** pencils. How many pencils altogether?

2. A classroom has **18** rows of chairs with **15** chairs in each row. How many chairs are there?

3. A bus has **47** seats. A school books **23** buses for a trip. How many seats are there in total?

4. Mr Bowen keeps **18** fish tanks. Every tank holds **25** rubber ducks. How many rubber ducks does he own?

Part 3 · Dividing – the multiplication backwards

F8 The same three numbers, four ways

Every multiplication you know is also two divisions. The first one is done for you.

Word help – the words that mean divide

share · split between	divide it up so everybody gets the same
goes into	“6 goes into 54 nine times” means $54 \div 6 = 9$
how many each	the answer to a division
left over · remainder	what will not divide up evenly

Multiplication	The other one	Division	The other one
$6 \times 9 = 54$	$9 \times 6 = 54$	$54 \div 6 = 9$	$54 \div 9 = 6$
$7 \times 8 = 56$			
		$72 \div 8 = \underline{\quad}$	
			$96 \div 12 = \underline{\quad}$

F9 Divide

- (a) $84 \div 7 = \underline{\quad}$ (c) $72 \div 6 = \underline{\quad}$ (e) $128 \div 4 = \underline{\quad}$ (g) $192 \div 8 = \underline{\quad}$
 (b) $96 \div 8 = \underline{\quad}$ (d) $135 \div 9 = \underline{\quad}$ (f) $161 \div 7 = \underline{\quad}$ (h) $216 \div 6 = \underline{\quad}$

Now four with something **left over**. Write them like this: $75 \div 8 = 9 \text{ r } 3$.

- (i) $58 \div 7 = \underline{\quad}$ (k) $85 \div 9 = \underline{\quad}$
 (j) $100 \div 6 = \underline{\quad}$ (l) $143 \div 4 = \underline{\quad}$

F10 Find the missing number

Every box holds one number. To find it, **divide**.

- (a) $6 \times \square = 54$ (c) $9 \times \square = 108$ (e) $12 \times \square = 60$
 (b) $\square \times 7 = 84$ (d) $\square \times 8 = 96$ (f) $\square \times 5 = 115$

F11 Share it out

1. **144** pencils are shared equally between **12** students. How many does each student get?

2. **90** students go on a trip. Each bus holds **16** students. How many buses are needed?
Careful: the last bus will not be full, and nobody can be left behind.

3. Mr Bowen shares **216** stickers equally between **8** classes. How many stickers does each class get?

4. Mr Bowen buys **175** rubber ducks and puts the same number into each of his **18** fish tanks. How many go into each tank? How many are left over?

Part 4 • The same grid, with a letter in it**F12 A letter is a number you do not know yet**

Collect the like terms. The first one is done for you.

Two rules, and that is all of it

Only the same letter can be added. $3a + 2a = 5a$, but $3a + 2b$ **cannot be added** and is already finished.

A letter on its own means one of it. $5x - x = 5x - 1x = 4x$, **not 5**.

(a) $3a + 2a = 5a$

(c) $7c - 2c = \underline{\hspace{2cm}}$

(e) $m + m = \underline{\hspace{2cm}}$

(b) $4b + 5b = \underline{\hspace{2cm}}$

(d) $5x - x = \underline{\hspace{2cm}}$

(f) $6y - y = \underline{\hspace{2cm}}$

(g) $2p + 3q =$ _____

(h) $8k - 3k =$ _____

(i) $4n + 7 =$ _____

F13 One box has a letter in it

Exactly the grid from Part 1. One box is a letter, so you cannot work that box out — **leave it as it is**.

$4(x + 3)$ is 4×16 with the 16 hidden

$\times$	x	3
4	$4x$	12

$4(x + 3) = 4x + 12$

$4 \times x$ is **$4x$** — you cannot finish it, so you write it and move on. 4×3 is 12. Then **add the boxes**, the same as always.

$\times$	x	5
2		

(a) $2(x + 5)$
 $=$ _____

$\times$	m	4
3		

(b) $3(m + 4)$
 $=$ _____

$\times$	y	2
6		

(c) $6(y + 2)$
 $=$ _____

$\times$	k	7
5		

(d) $5(k + 7)$
 $=$ _____

F14 Fill the boxes

No grid. Write the two numbers straight into the boxes.

(a) $3(a + 2) = \square a + \square$

(d) $2(d + 9) = \square d + \square$

(b) $5(b + 4) = \square b + \square$

(e) $4(e + 6) = \square e + \square$

(c) $7(c + 1) = \square c + \square$

(f) $8(f + 3) = \square f + \square$

F15 A minus inside

The minus sign belongs to the number after it, and it goes into the brackets with that number.

Same grid, one sign

$3(x - 2) = 3x - 6$

$5(y - 4) = 5y - 20$

(a) $4(x - 3) =$ _____

(d) $9(n - 1) =$ _____

(b) $6(m - 2) =$ _____

(e) $3(p - 5) =$ _____

(c) $2(k - 7) =$ _____

(f) $7(q - 2) =$ _____

F16 Find the number outside

This is **F10** again, with a bracket around it. Divide, exactly as you did there.

(a) $\square (x + 2) = 5x + 10$

(d) $\square (k + 5) = 6k + 30$

(b) $\square (y + 3) = 4y + 12$

(e) $\square (p + 4) = 3p + 12$

(c) $\square (m + 1) = 7m + 7$

(f) $\square (n + 6) = 8n + 48$

F17 Check it with a number**This one proves the whole sheet**

You wrote $4(x + 3) = 4x + 12$ in F13. Now **put $x = 10$ into both sides** and see whether they agree.

(a) The **left** side. $x + 3$ is $10 + 3 = \square$, so $4(x + 3)$ is $4 \times \square = \square$

(b) The **right** side. $4x$ is $4 \times 10 = \square$, so $4x + 12$ is $\square + 12 = \square$

(c) Did the two sides agree? _____

(d) You have just worked out 4×13 twice, in two different ways. Which two numbers did you add together the second time? _____

(e) Now do the whole thing again on your own, with $3(m + 4)$ and $m = 20$.

(i) Expand it. _____

(ii) The **left** side: work out $m + 4$ first, then multiply by 3. _____

(iii) The **right** side: put $m = 20$ into your expanded answer. _____

(iv) Did they agree? _____